Yifei Jin

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EDUCATION

Ph.D. in Electrical & Computer Engineering 08/2021 – Expected 05/2026
University of Illinois at Urbana-Champaign (UIUC), Urbana, IL
B.S. in Electrical & Computer Engineering 08/2016 – 05/2019
Worcester Polytechnic Institute (WPI), Worcester, MA

WORK EXPERIENCE

Display Module Design Engineer, Apple Inc., Cupertino, CA 03/2026 – Present
- Defining display module EE architecture and features
- Designing display electronic components including flexible PCBs and SiPs
- Collaborating with vendors on design execution/validation, and cross-functional teams on system integration
- Define electro-optical performance metrics through extensive measurements and modeling of hardware

Display Electrical Engineer Intern, Apple Inc., Cupertino, CA 05/2025 – 08/2025
- Improved OLED panel front-of-screen quality and lowered silicon cost by developing novel uniformity compensation methods
- Designed a TFT physics-based non-linear interpolation model for diverse operating conditions
- Collaborated with Module, Optics, and Silicon teams to align specs, build lab setups, and analyze results
- Led comprehensive brightness/color characterization guiding algorithm validation and design improvements

Graduate Teaching Assistant, UIUC 08/2023 – 12/2025
- Taught digital circuit design through FPGA, including Verilog coding and SPI/I²C protocols
- Mentored ~120 students on FPGA-based camera systems with real-time object tracking

Hardware Engineer, Teradyne Inc., North Reading, MA 06/2019 – 04/2021
- Developed high-voltage VI channel modules for power semiconductor DC testing on UltraFLEX*plus* ATE
- Executed circuit bring-up/verification in VB, diagnosing and resolving circuit defects that improved stability and performance under demanding test conditions

RESEARCH PROJECTS

Biomimetic Image Sensor for Metastatic Lymph Node Detection, UIUC 01/2022 – 12/2024
- Co-designed a UV-Color-NIR camera system, enhancing image-guided surgery and pathology capabilities
- Characterized sensor QE, dynamic range, and uniformity; implemented full ISP pipeline including 3A algorithms, color correction, and gamma correction
- Validated on 94 surgical samples with 97% sensitivity and 89% specificity for metastatic lymph node detection

In-vivo UV-Color-NIR Lensless 3D Microscope, UIUC 01/2024 – Present
- Enabled single-shot 3D fluorescence imaging in vivo by designing a multispectral lensless microscope spanning UV, color and NIR
- Achieved ~30 µm resolution comparable to a 4× microscope through computational imaging algorithms
- Integrated multi-wavelength fluorescence excitation sources via PCB for compact, surgical-ready hardware

CNN-based Demosaicing for Color-NIR Sensors, UIUC 05/2023 – 03/2024
- Developed real-time residual CNN for demosaicing hexachromatic (Color-NIR) sensor mosaics
- Achieved ~3 dB PSNR gain over baseline on clinical datasets; published in *Journal of Biomedical Optics*

SKILLS

- Programming: Python, PyTorch, OpenCV, MATLAB, Verilog/VHDL, C/C++, Visual Basic
- Hardware: FPGA, PCB design, SPI/I²C protocols, Oscilloscope, Function Generator, Soldering
- Optical/Display Characterization: Monochromator, Integrating Sphere, Optical Power Meter, Spectroscopy, Display Color Analyzer, High-resolution imaging colorimeter, and Optomechanics
- Software: Vivado, Quartus II, Cadence, KiCad, Simulink, Linux, Code Composer Studio